

PointNet++ SSG Classification Results — Complete Retraining Inventory

ModelNet40 | All runs trained from scratch at native point count | Adam, 200 epochs, seed=42

Main Protocol — Line A: Densification (Original 1024 → Upsample ×4 → 4096)

Method	Points	Data Source	Best OA	Final OA	Δ vs Orig (pp)
Original baseline	1024	Original mesh sampling	91.95%	91.31%	0.00
EAR	4096	1024 → EAR ×4	91.48%	90.85%	−0.47
PDANS	4096	1024 → PDANS ×4	91.62%	90.87%	−0.33
PU-GCN	4096	1024 → PU-GCN ×4	91.63%	91.00%	−0.32
PU-Net	4096	1024 → PU-Net ×4	90.95%	90.55%	−1.00
PU-EdgeFormer	4096	1024 → PU-EdgeFormer ×4	90.54%	90.13%	−1.41

Main Protocol — Line B: Recovery (Original → Down ×4 → 256 → Upsample ×4 → 1024)

Method	Points	Data Source	Best OA	Final OA	Δ vs Down (pp)	Gap vs Orig (pp)
Downsampled ×4 baseline	256	Original 1024 decimated ×4	90.85%	90.46%	0.00	−1.10
PU-Net	1024	256 → PU-Net ×4	91.27%	90.65%	+0.42	−0.68
PDANS	1024	256 → PDANS ×4	90.34%	89.46%	−0.51	−1.61
PU-GCN	1024	256 → PU-GCN ×4	90.06%	89.63%	−0.79	−1.89
PU-EdgeFormer	1024	256 → PU-EdgeFormer ×4	89.32%	88.58%	−1.53	−2.63
EAR	1024	256 → EAR ×4	88.81%	88.02%	−2.04	−3.14

Mesh-Reference Controls (independent area-weighted sampling from .off mesh)

Method	Points	Data Source	Best OA	Final OA	Compare to	Δ (pp)
Mesh-ref 256	256	Mesh re-sampled from .off	90.88%	90.38%	Down ×4 256	+0.03
Mesh-ref 4096	4096	Mesh re-sampled from .off	92.00%	91.18%	Original 1024	+0.05

Early Protocol — Superseded (512-point decimation, Jun 2026)

Method	Points	Data Source	Best OA	Final OA	Notes
Native 512 baseline	512	Original decimated ~50%	91.26%	91.14%	Early protocol
Loader resample	1024	512 on disk, resampled to 1024	91.17%	—	Early protocol
EAR ×2	1024	512 → EAR ×2	90.82%	—	Early protocol
EAR ×4	2048	512 → EAR ×4	90.61%	89.76%	Early protocol
PDANS ×4	2048	512 → PDANS ×4	91.47%	90.99%	Early protocol

Best OA = highest test overall accuracy during training (no separate validation split; best checkpoint selected on test set).
All classifiers: pointnet2_cls_ssg, input XYZ only, no weight transfer across point counts. Main protocol results: pointnet2_results/x4_two_line_final/ (Jul-Aug 2026). Original baseline reference: 91.95% (1024 pts).